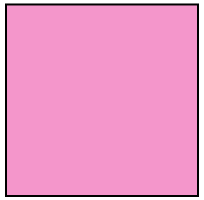
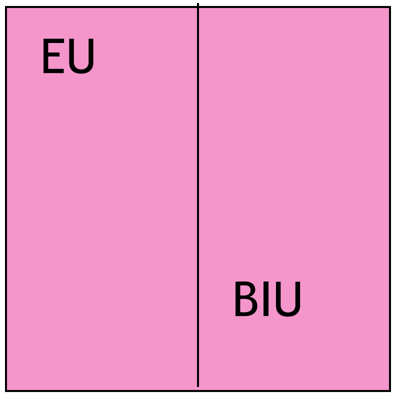
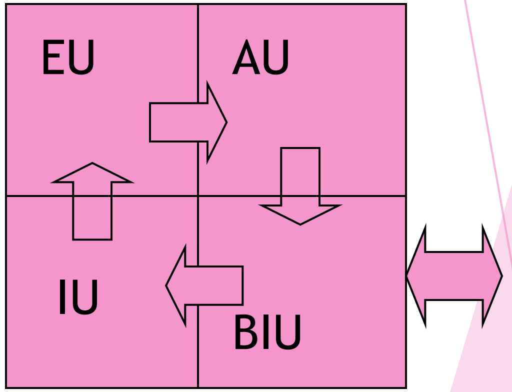
Internal Block Diagram of 80286



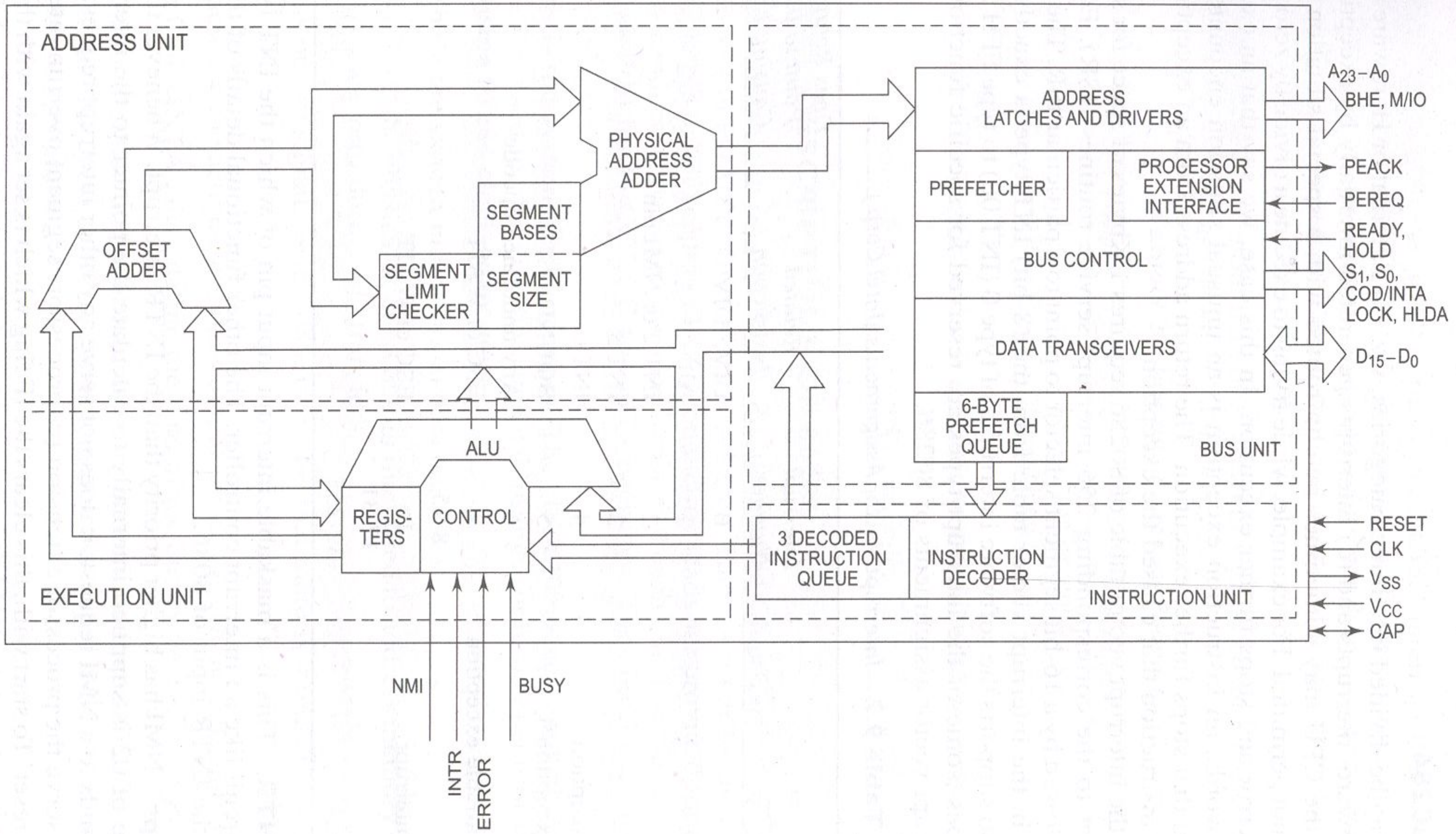
8085



8086



80286


ADDRESS UNIT
PHYSICAL ADDRESS ADDER
SEGMENT BASES
OFFSET ADDER
SEGMENT LIMIT CHECKER
SEGMENT SIZE
ADDRESS LATCHES AND DRIVERS
$A_{23}-A_0$
BHE, M/IO
PREFETCHER
PROCESSOR EXTENSION INTERFACE
PEACK
PEREQ
READY, HOLD
S_1, S_0, COD/INTA LOCK, HLDA
BUS CONTROL
DATA TRANSCEIVERS
$D_{15}-D_0$
6-BYTE PREFETCH QUEUE
BUS UNIT
ALU
REGIS- TERS
CONTROL
3 DECODED INSTRUCTION QUEUE
INSTRUCTION DECODER
INSTRUCTION UNIT
EXECUTION UNIT
RESET
CLK
V_{SS}
V_{CC}
CAP
NMI
INTR
ERROR
BUSY

Functional Parts

1. Address unit
2. Bus unit
3. Instruction unit
4. Execution unit

Address Unit

- Calculate the physical addresses of the instruction and data that the CPU want to access
- Address lines derived by this unit may be used to address different peripherals.
- Physical address computed by the address unit is handed over to the BUS unit.

Bus Unit

- Performs all memory and I/O read and write operations.
- Take care of communication between CPU and a coprocessor.
- Transmit the physical address over address bus $A_0 – A_{23.}$
- Prefetcher module in the bus unit performs this task of prefetching.
- Bus controller controls the prefetcher module.
- Fetched instructions are arranged in a 6 – byte prefetch queue.

Instruction Unit

- Receive arranged instructions from 6 byte prefetch queue.
- Instruction decoder decodes up to 3 prefetched instruction and are latched them onto a decoded instruction queue.
- Output of the decoding circuit drives a control circuit in the Execution unit.

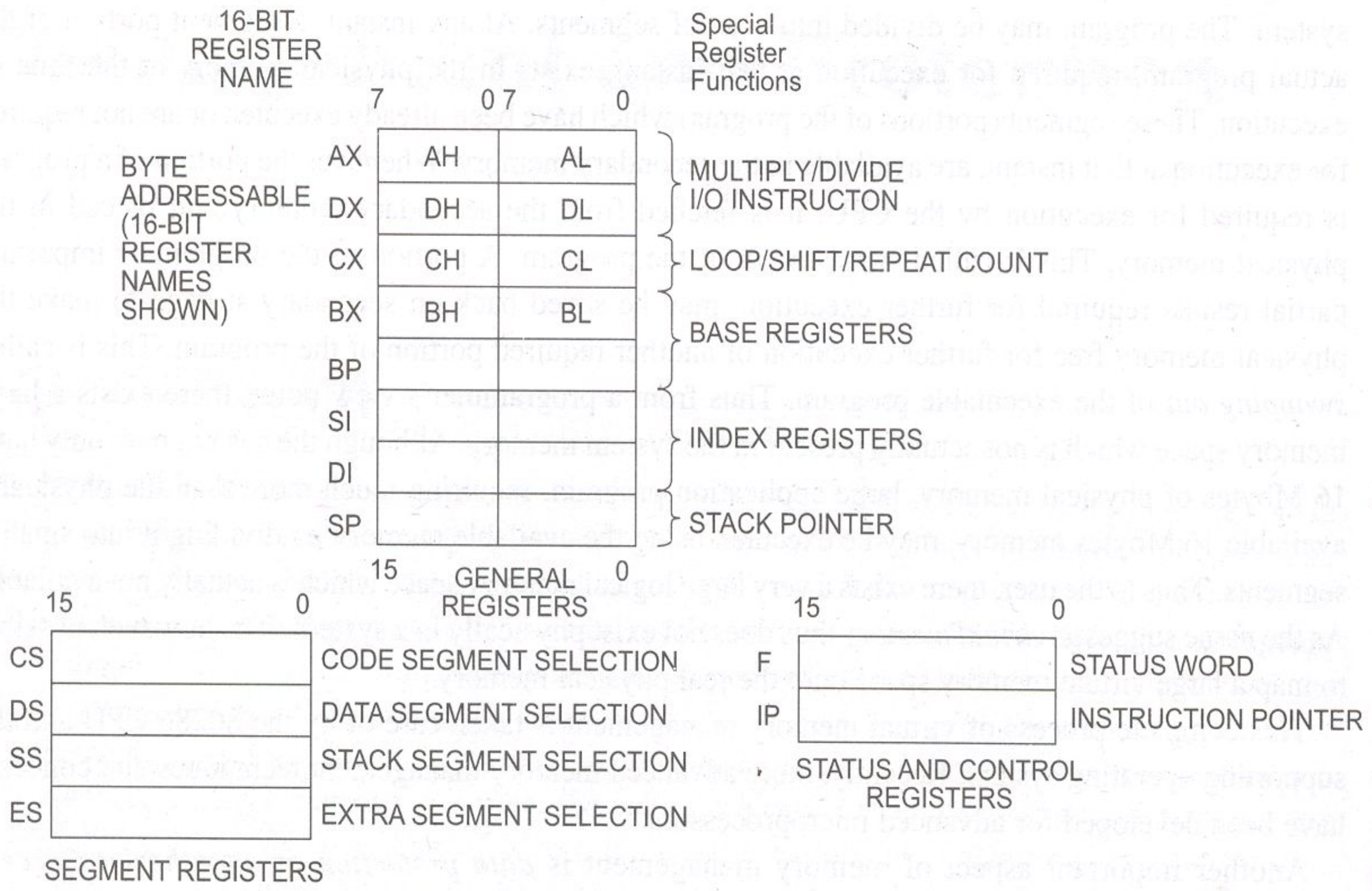
Execution unit

- EU executes the instructions received from the decoded instruction queue sequentially.
- Contains Register Bank.
- contains one additional special register called **Machine status word (MSW)** register --- lower 4 bits are only used.
- ALU is the heart of execution unit.
- After execution ALU sends the result either over data bus or back to the register bank.

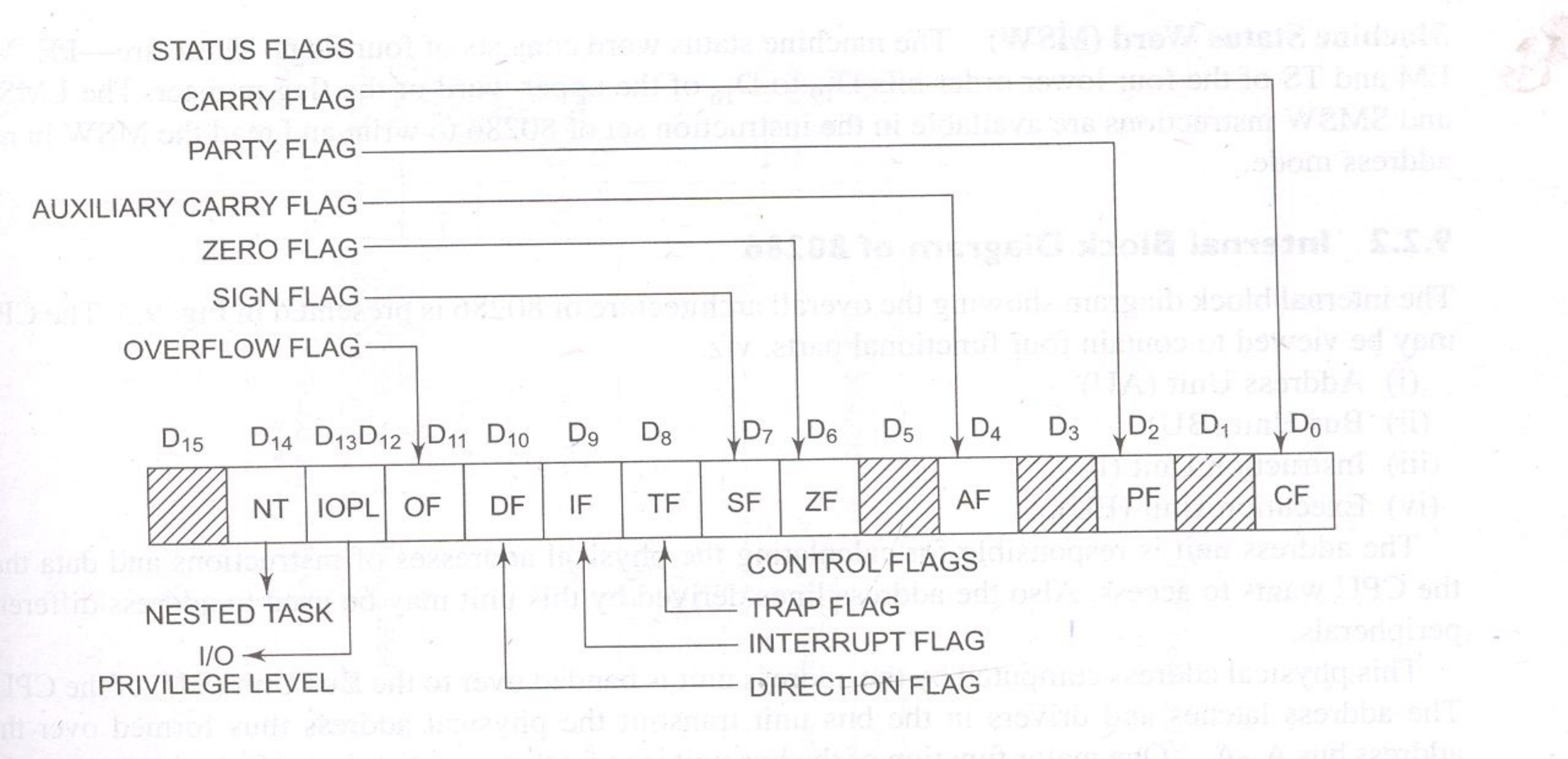
Register organization of 80286

The 80286 CPU contains the same set of registers, as in 8086.

1. Eight 16-bit general purpose registers.
2. Four 16 bit segment registers.
3. Status and control register.
4. Instruction pointer.


16-BIT REGISTER NAME
Special Register Functions
7
0 7
0
AX
AH
AL
BYTE ADDRESSABLE (16-BIT REGISTER NAMES SHOWN)
DX
DH
DL
CX
CH
CL
BX
BH
BL
BP
SI
DI
SP
MULTIPLY/DIVIDE
I/O INSTRUCTON
LOOP/SHIFT/REPEAT COUNT
BASE REGISTERS
INDEX REGISTERS
STACK POINTER
15
0
GENERAL REGISTERS
15
0
CS
CODE SEGMENT SELECTION
DS
DATA SEGMENT SELECTION
SS
STACK SEGMENT SELECTION
ES
EXTRA SEGMENT SELECTION
SEGMENT REGISTERS
15
0
F
STATUS WORD
IP
INSTRUCTION POINTER
STATUS AND CONTROL REGISTERS

Flag Registers

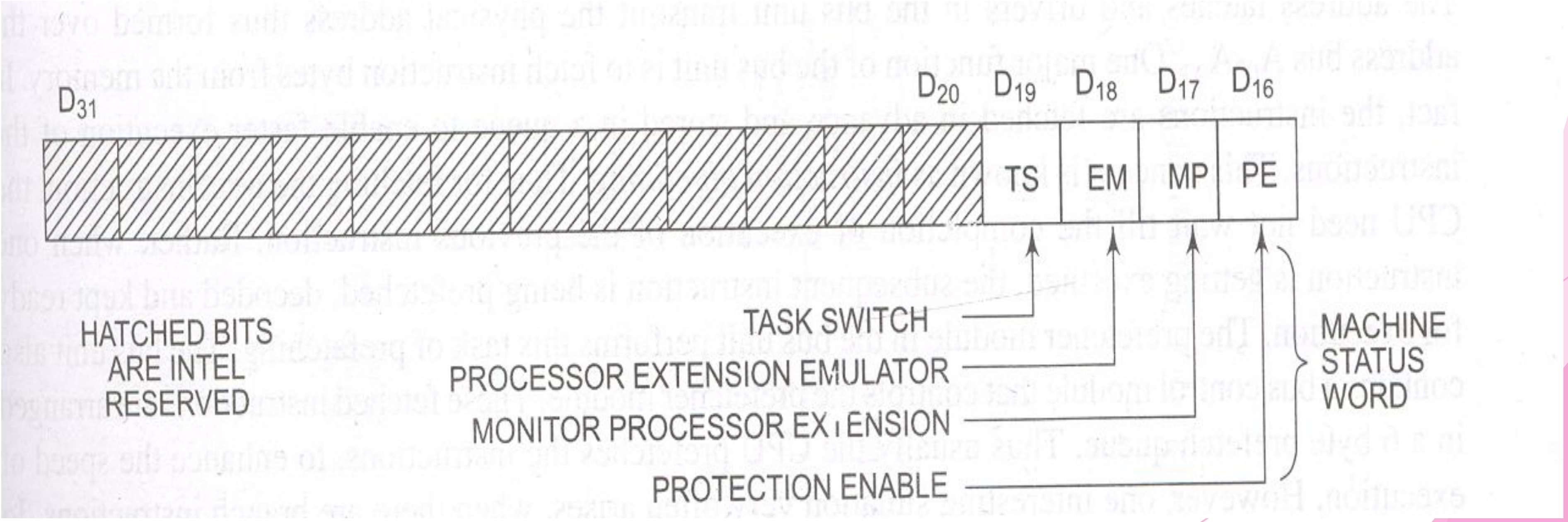

STATUS FLAGS
CARRY FLAG
PARTY FLAG
AUXILIARY CARRY FLAG
ZERO FLAG
SIGN FLAG
OVERFLOW FLAG
D15 D14 D13 D12 D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0
NT IOPL OF DF IF TF SF ZF AF PF CF
CONTROL FLAGS
TRAP FLAG
INTERRUPT FLAG
DIRECTION FLAG
NESTED TASK
I/O
PRIVILEGE LEVEL

- IOPL – Input Output Privilege Level flags (bit D12 and D13)
  - IOPL is used in protected mode operation to select the privilege level for I/O devices. IF the current privilege level is higher or more trusted than the IOPL, I/O executed without hindrance. If the IOPL is lover than the current privilege level, an interrupt occurs, causing execution to suspend. Note that IPOL 00 is the highest or more trusted; and IOPL 11 is the lowest or least trusted.
- NT – Nested task flag (bit D14)
  - When set, it indicates that one system task has invoked another through a CALL instruction as opposed to a JMP. For multitasking this can be manipulated to our advantage

Machine Status Word

- Consist of four flags. These are – PE, MP, EM and TS

- LMSW & SMSW instruction are available in the instruction set of 80286 to write and read the MSW in real address mode.



- PE - Protection enable
  - Protection enable flag places the 80286 in protected mode, if set. This can only be cleared by resetting the CPU.
- MP – Monitor processor extension
  - flag allows WAIT instruction to generate a processor extension.
- EM – Emulate processor extension flag,
  - if set , causes a processor extension absent exception and permits the emulation of processor extension by CPU.
- TS – Task switch
  - if set this flag indicates the next instruction using extension will generate exception 7, permitting the CPU to test whether the current processor extension is for current task.

Additional Instructions of Intel 80286

Sl no	Instruction	Purpose
1.	CLTS	Clear the task – switched bit
2.	LDGT	Load global descriptor table register
3.	SGDT	Store global descriptor table register
4.	LIDT	Load interrupt descriptor table register
5.	SIDT	Store interrupt descriptor table register
6.	LLDT	Load local descriptor table register
7.	SLDT	Store local descriptor table register
8.	LMSW	Load machine status register
9.	SMSW	Store machine status register

Sl no	Instruction	Purpose
10.	LAR	Load access rights
11.	LSL	Load segment limit
12.	SAR	Store access right
13.	ARPL	Adjust requested privilege level
14.	VERR	Verify a read access
15.	VERW	Verify a write access

- **CLTS**
  - The clear task - switched flag instruction clears the TS (Task - switched) flag bit to a logic 0.
- **LAR**
  - The load access rights Instruction reads the segment descriptor and place a copy of the access rights byte into a 16 bit register.
- **LSL**
  - The load segment limit instruction Loads a user - specified register with the segment limit.

- **VERR**
  - The verify for read access instruction verifies that a segment can de read.
- **VERW**
  - The verify for write access instruction is used to verify that a segment can be written.
- **ARPL**
  - The Adjust request privilege level instruction is used to test a selector so that the privilege level of the requested selector is not violated.